PUSAN NATIONAL UNIVERSITY

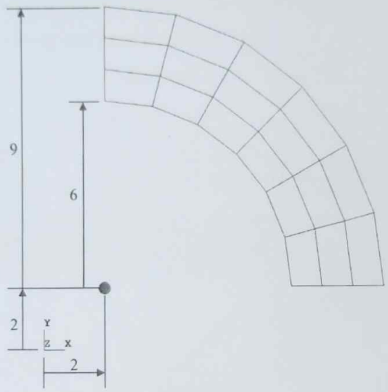
Dept. of Aerospace Engineering

Computational Structural Analysis

Mid-exam. Oct. 21, 2025

Problem 1

Create the finite element model, as shown in below figure, using a local cylindrical coordinate system, NGEN and EGEN commands. Use PLANE182 as an element type.

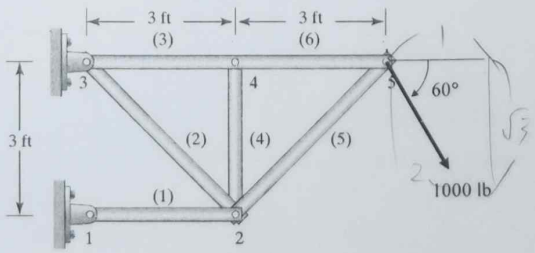


Problem 2

Consider the balcony truss shown in the figure below, where all members are made of Douglas-fir wood with a modulus of elasticity of $E = 1.90 \times 10^6$ lb/in^2 and a cross-sectional area of 8 in^2. Determine the axial stress in each member and the maximum nodal displacement. Trigonometric functions should not be used to apply concentrated forces at position 5. Instead, use the following approaches,

(a) NROTAT command.

(b) NMODIF command.

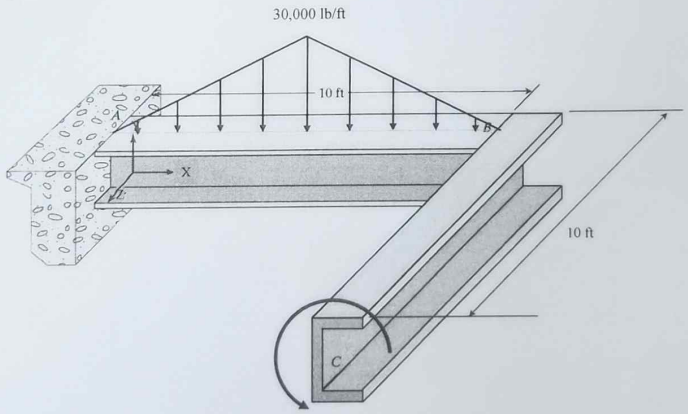


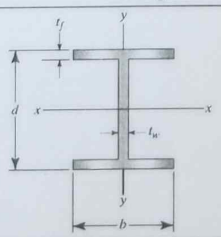
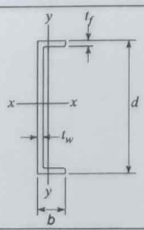
Problem 3

A beam, fixed at one end in concrete(point A), is subjected to gradually varying distributed load. The beam is made of steel ($E = 30 \times 10^6$ lb/in^2, $\nu = 0.3$) and consists of a W8×67 wide-flange section and a C9×20 channel section, whose geometric dimensions are listed in the table below. Use the SECOFFS command to align the wide-flange and channel sections so that they fit at point *B*.

(a) Find the maximum bending stress and the angle of twist at point *C*.

(b) How much counter-twisting moment should be applied to eliminate the twist about the Z-axis, i.e., to ensure zero angle of twist at point *C*?



Designation		d	t_w	b	t_f
	W8×67	9 in	0.570 in	8.280 in	0.935 in
	C9×20	9 in	0.448 in	2.648 in	0.413 in